

On hyperbolic approximations for a class of dispersive and diffusive-dispersive equations

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Joint work with
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The problem PDE

We consider for some interval $I \subset \mathbb{R}$ the scalar initial-value problem

$$\begin{aligned} u_t + f(u)_x &= \varepsilon u_{xx} + \gamma u_{xxx} \quad \text{in } I_T := I \times (0, T), \quad T > 0, \\ u(\cdot, 0) &= u_0(x). \end{aligned}$$

where

- $u(x, t)$ is the unknown.
 - $f : \mathbb{R} \rightarrow \mathbb{R}$ is a smooth flux function.
 - $\gamma \in \mathbb{R}, \varepsilon \geq 0$.
- ⇒ Contains several models of wide interest: KdV, mKdV, KdV-Burgers, mKdV-Burgers, Gardner equation, etc.
- ⇒ Toy model to describe phase transition dynamics when viscous and capillary effects are taken into account.

The problem PDE - purely dispersive case

We consider for some interval $I \subset \mathbb{R}$ the scalar initial-value problem

$$\begin{aligned} u_t + f(u)_x &= \gamma u_{xxx} \quad \text{in } I_T := I \times (0, T), \quad T > 0, \\ u(\cdot, 0) &= u_0(x). \end{aligned}$$

$$\begin{aligned} \text{KdV eq:} \quad & f(u) = u^2/2, \quad \gamma \neq 0 \quad \Rightarrow \quad u_t + uu_x + \gamma u_{xxx} = 0, \\ \text{mKdV eq:} \quad & f(u) = u^3, \quad \gamma \neq 0 \quad \Rightarrow \quad u_t + 3u^2 u_x + \gamma u_{xxx} = 0, \\ \text{Gardner eq:} \quad & f(u) = u^3 + \mu u^2, \quad \gamma < 0 \quad \Rightarrow \quad u_t + (3u^2 + 2\mu u) u_x + \gamma u_{xxx} = 0. \end{aligned}$$

- Smooth solitary waves
- Dispersive shockwaves in the low dispersion limit $\gamma \rightarrow 0$.

The problem PDE - dissipative/dispersive case

We consider for some interval $I \subset \mathbb{R}$ the scalar initial-value problem

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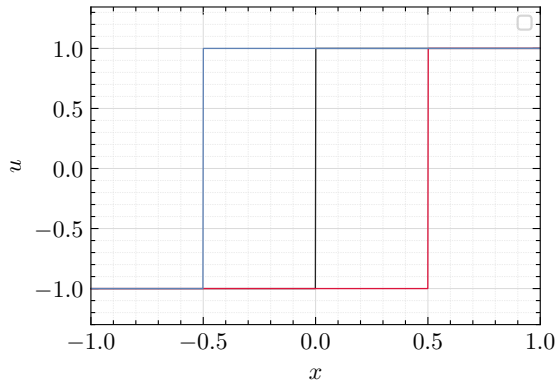
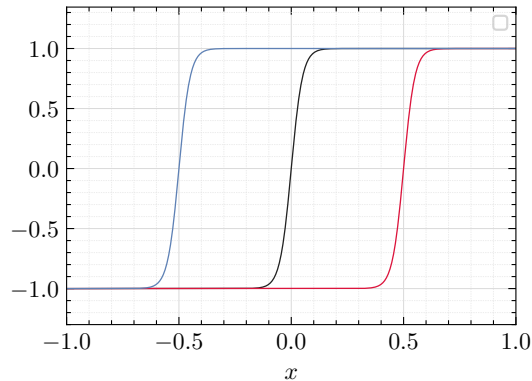
$$u(\cdot, 0) = u_0(x).$$

KdV-B eq: $f(u) = u^2/2, \quad \varepsilon > 0, \quad \gamma \neq 0 \quad \Rightarrow \quad u_t + uu_x + \gamma u_{xxx} = \varepsilon u_{xx},$

mKdV-B eq: $f(u) = u^3, \quad \varepsilon > 0, \quad \gamma \neq 0 \quad \Rightarrow \quad u_t + 3u^2 u_x + \gamma u_{xxx} = \varepsilon u_{xx}.$

- Smooth solitary waves
- Possible undercompressive shocks in the sharp-interface limit.

Undercompressive shocks as limits



Objectives

We would like to rewrite/approximate the equation

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- I. The dissipationless part comes from a variational principle.
- II. The new model admits an entropy.
- III. It can be solved numerically fairly easily.
- IV. The same numerical method addresses most aspects/regimes of the PDE.
- V. Well-posed at least for a class of initial data (whenever possible).

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First-order hyperbolic equations

- locally well-posed IVP (Symmetric Hyperbolic)
- A very rich literature on numerical methods.
- Bounded wave speeds (Principle of causality)

Plan of presentation

- 1 The hyperbolic approximation
- 2 Model properties
- 3 Numerical results

Non-exhaustive references on hyperbolic approximations

Heat Cattaneo, Carlo. "Sulla conduzione del calore." Atti Sem. Mat. Fis. Univ. Modena 3 (1948): 83-101.

$$T_t - \nabla \cdot (D \nabla T) = 0, \quad \Rightarrow \begin{cases} T_t - \nabla \cdot \mathbf{J} = 0 \\ \varepsilon \mathbf{J}_t - D \nabla T = -\mathbf{J}, \quad 0 < \varepsilon \ll 1 \end{cases}$$

Heat GK Model [Guyer-Krumhantsl 1964], Heat transfer in compressible flows [Dhaouadi, Gavriluk 2024], etc.

NSE GPR Model [Dumbser et al. 2016], GENERIC Framework [Peshkov et al. 2018] , [Nishikawa et al. 2016], etc

⇒ Other diffusive PDEs:

CH Cahn-Hilliard equations [Dhaouadi et al. 2025]

Example references on hyperbolic approximations for dispersive PDEs

- 2008 **V. Yu. Liapidevskii and K. N. Gavrilova.** *Dispersion and blockage effects in the flow over a sill.* Journal of Applied Mechanics and Technical Physics, 49(1).
- 2017 **N. Favrie and S. Gavrilyuk.** *A rapid numerical method for solving Serre–Green–Naghdi equations describing long free surface gravity waves.* Nonlinearity, 30(7), pp. 2718–2736.
- 2019 **F. Dhaouadi, N. Favrie and S. Gavrilyuk.** *Extended Lagrangian approach for the defocusing nonlinear Schrödinger equation.* Studies in Applied Mathematics, 142(3), pp. 336–358.
- 2022 **C. Besse, S. Gavrilyuk, M. Kazakova, P. Noble.** *Perfectly matched layers methods for mixed hyperbolic–dispersive equations.* Water Waves, 4(3), pp. 313–343.
- 2025 **D. I. Ketcheson and A. Biswas.** *Approximation of arbitrarily high-order PDEs by first-order hyperbolic relaxation.* Nonlinearity, 38(5), 055002.
- (Works of Bresch, Noble, Vila, Romenski, Dumbser, etc.)

Lower order approximations of the target PDE

$$u_t + f(u)_x = \gamma u_{xxx} + \varepsilon u_{xx}$$

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- Second-order elliptic relaxation [Corli & Rohde 2012]

$$\begin{aligned} u_t^\alpha + f(u^\alpha)_x &= -\alpha (u^\alpha - c^\alpha)_x + \varepsilon u_{xx}^\alpha, \\ -\gamma c_{xx}^\alpha &= \alpha (u^\alpha - c^\alpha), \end{aligned} \quad 1 \ll \alpha.$$

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- Second-order parabolic relaxation [Corli, Rohde & Schleper 2014]

$$\begin{aligned} u_t^\alpha + f(u^\alpha)_x &= -\alpha (u^\alpha - c^\alpha)_x + \varepsilon u_{xx}^\alpha, \\ \beta c_t^\alpha - \gamma c_{xx}^\alpha &= \alpha (u^\alpha - c^\alpha), \quad 0 < \beta \ll 1. \end{aligned}$$

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- Can we apply a simple wave-type relaxation ?

$$\begin{aligned} u_t^\alpha + f(u^\alpha)_x &= -\alpha (u^\alpha - c^\alpha)_x + \varepsilon u_{xx}^\alpha, \\ \beta \textcolor{red}{c}_{tt}^\alpha - \gamma c_{xx}^\alpha &= \alpha (u^\alpha - c^\alpha). \end{aligned}$$

The main idea

- The only part we want hyperbolic is the purely dispersive system

$$u_t + f(u)_x = \gamma u_{xxx} \quad \text{in } I_T := I \times (0, T), \quad T > 0,$$
$$u(\cdot, 0) = u_0(x).$$

- The parabolic term is to remain as it is.

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- 1 Build on the proposed wave-type approximation in the purely dispersive case

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$$u_t^\alpha + f(u^\alpha)_x = -\alpha (u^\alpha - c^\alpha)_x,$$

$$\beta c_{tt}^\alpha - \gamma c_{xx}^\alpha = \alpha (u^\alpha - c^\alpha).$$

- 2 Extend to the case with parabolic dissipative term εu_{xx} .

First order hyperbolic relaxation

1 Start from

$$\begin{aligned}u_t^\alpha + f(u^\alpha)_x &= -\alpha (u^\alpha - c^\alpha)_x, \\ \beta c_{tt}^\alpha - \gamma c_{xx}^\alpha &= \alpha (u^\alpha - c^\alpha).\end{aligned}$$

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Then we obtain

$$\begin{aligned} u_t^\alpha + f(u^\alpha)_x + \alpha (u^\alpha - c^\alpha)_x &= 0, & u_t^\alpha + f(u^\alpha)_x + \psi_x^\alpha &= 0, \\ \beta w_t^\alpha - \gamma p_x^\alpha &= \alpha (u^\alpha - c^\alpha), & \Rightarrow \quad \beta w_t^\alpha - \gamma p_x^\alpha &= \psi^\alpha, \\ c_t^\alpha &= w^\alpha, & \psi_t^\alpha + \alpha f(u^\alpha)_x + \alpha \psi_x &= -\alpha w^\alpha, \\ p_t^\alpha - w_x^\alpha &= 0. & p_t^\alpha - w_x^\alpha &= 0. \end{aligned}$$

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$\Rightarrow$

$$\begin{aligned}u_t^\alpha + f(u^\alpha)_x + \psi_x &= 0, \\ \psi_t^\alpha + \alpha f(u^\alpha)_x + \alpha \psi_x &= -\alpha w^\alpha, \\ \beta w_t^\alpha - \gamma p_x^\alpha &= \psi, \\ p_t^\alpha - w_x^\alpha &= 0.\end{aligned}$$

Hyperbolicity

We write the homogeneous system in quasilinear form

$$\mathbf{U}_t^\alpha + \mathbf{Df}(\mathbf{U}^\alpha)\mathbf{U}_x^\alpha = 0.$$

where the Jacobian is given by

$$\mathbf{Df}(\mathbf{U}) = \begin{pmatrix} f'(u) & 1 & 0 & 0 \\ \alpha f'(u) & \alpha & 0 & 0 \\ 0 & 0 & 0 & -\gamma/\beta \\ 0 & 0 & -1 & 0 \end{pmatrix}.$$

The eigenvalues of $\mathbf{Df}(\mathbf{U})$ compute as the real numbers $\lambda_1, \dots, \lambda_4$ given by

$$\lambda_1 = -\sqrt{\gamma/\beta}, \quad \lambda_2 = 0, \quad \lambda_3 = \sqrt{\gamma/\beta}, \quad \lambda_4 = f'(u) + \alpha.$$

$\Rightarrow$ System is hyperbolic

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$\Rightarrow$ System is hyperbolic **only if** $\gamma/\beta > 0$

The mistake...

... Comes from the very beginning

$$\begin{aligned}u_t^\alpha + f(u^\alpha)_x &= -\alpha (u^\alpha - c^\alpha)_x, \\ \beta c_{tt}^\alpha - \gamma c_{xx}^\alpha &= \alpha (u^\alpha - c^\alpha).\end{aligned}$$

The second equation is a wave-type operator only if β and γ are of same signs!

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The fix

$$\begin{aligned}u_t^\alpha + f(u^\alpha)_x &= -\operatorname{sgn}(\gamma) \alpha (u^\alpha - c^\alpha)_x, \\ \beta c_{tt}^\alpha - |\gamma| c_{xx}^\alpha &= \alpha (u^\alpha - c^\alpha).\end{aligned}$$

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We call the wave velocity $v_\beta = \sqrt{|\gamma|/\beta}$

1 - Fixed hyperbolicity

Now the system writes as

$$\begin{aligned} u_t^\alpha + (f(u^\alpha) + \operatorname{sgn}(\gamma)\psi^\alpha)_x &= 0, \\ \psi_t^\alpha + \alpha (f(u^\alpha) + \operatorname{sgn}(\gamma)\psi^\alpha)_x &= -\alpha w^\alpha, \\ w_t^\alpha - v_\beta^2 p_x^\alpha &= \beta^{-1} \psi^\alpha, \quad v_\beta = \sqrt{|\gamma|/\beta} \\ p_t^\alpha - w_x^\alpha &= 0 \end{aligned}$$

the Jacobian is now given by

$$\mathbf{Df}(\mathbf{U}) = \begin{pmatrix} f'(u) & \operatorname{sgn}(\gamma) & 0 & 0 \\ \alpha f'(u) & \alpha \operatorname{sgn}(\gamma) & 0 & 0 \\ 0 & 0 & 0 & -v_\beta^2 \\ 0 & 0 & -1 & 0 \end{pmatrix}.$$

The eigenvalues of $\mathbf{Df}(\mathbf{U})$ compute as the real numbers $\lambda_1, \dots, \lambda_4$ given by

$$\lambda_1 = -v_\beta, \quad \lambda_2 = 0, \quad \lambda_3 = v_\beta, \quad \lambda_4 = f'(u) + \alpha \operatorname{sgn}(\gamma).$$

Least action principle - original dispersive PDE

The equation

$$u_t + f(u)_x = \gamma u_{xxx}$$

admits the following Lagrangian

$$\mathcal{L}[\eta_t, \eta_x, \eta_{xx}] := \int_I \left(\operatorname{sgn}(\gamma) \left(\frac{1}{2} \eta_x \eta_t + F(\eta_x) \right) + \frac{|\gamma|}{2} (\eta_{xx})^2 \right) dx,$$

where $u(x, t) = \eta_x(x, t)$.

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where $u(x, t) = \eta_x(x, t)$. Simply apply Hamilton's principle for $t_0, t_1 \in [0, T]$ to the action

$$\mathcal{S}[\eta_t, \eta_x, \eta_{xx}] = \int_{t_0}^{t_1} \mathcal{L}[\eta_t, \eta_x, \eta_{xx}] dt.$$

to obtain the Euler-Lagrange equation

$$\left(\frac{\partial \mathcal{L}}{\partial \eta_t} \right)_t + \left(\frac{\partial \mathcal{L}}{\partial \eta_x} \right)_x - \left(\frac{\partial \mathcal{L}}{\partial \eta_{xx}} \right)_{xx} = \operatorname{sgn}(\gamma) (\eta_{xt} + (f(\eta_x))_x) - |\gamma| \eta_{xxxx} = 0.$$

2- Least action principle - hyperbolic PDE

Consider the modified Lagrangian (again $u(x, t) = \eta_x(x, t)$)

$$\mathcal{L}_\alpha[\eta_t, \eta_x, c, c_t, c_x] = \int_I \left(\operatorname{sgn}(\gamma) \left(\frac{1}{2} \eta_x \eta_t + F(\eta_x) \right) + \frac{|\gamma| c_x^2}{2} + \frac{\alpha}{2} (\eta_x - c)^2 - \frac{\beta}{2} c_t^2 \right) dx,$$

Hamilton's principle yields the two Euler-Lagrange equations

$$\begin{aligned} \left(\frac{\partial \mathcal{L}_\alpha}{\partial \eta_t} \right)_t + \left(\frac{\partial \mathcal{L}_\alpha}{\partial \eta_x} \right)_x &= \operatorname{sgn}(\gamma) (\eta_{xt} + f(\eta_x)_x) + \alpha (\eta_x - c)_x = 0, \\ -\frac{\partial \mathcal{L}_\alpha}{\partial c} + \left(\frac{\partial \mathcal{L}_\alpha}{\partial c_t} \right)_t + \left(\frac{\partial \mathcal{L}_\alpha}{\partial c_x} \right)_x &= -\beta c_{tt} + |\gamma| c_{xx} + \alpha (\eta_x - c) = 0, \\ \implies u_t + f(u)_x &= \operatorname{sgn}(\gamma) \alpha (u - c)_x, \quad \text{and} \quad \beta c_{tt} - |\gamma| c_{xx} = \alpha (u - c) \end{aligned}$$

3 - Entropy / Entropy-flux pair

Lemma (Entropy/entropy-flux pair)

The pair (E^α, Q^α) with

$$E^\alpha(\mathbf{U}) = \operatorname{sgn}(\gamma)F(u) + \frac{|\gamma|}{2}p^2 + \frac{1}{2\alpha}\psi^2 + \frac{\beta}{2}w^2, \quad (2)$$

and

$$Q^\alpha(\mathbf{U}) = \frac{1}{2} \operatorname{sgn}(\gamma) f(u)^2 + f(u)\psi + \operatorname{sgn}(\gamma) \frac{\psi^2}{2} - |\gamma|pw \quad (3)$$

is an entropy/entropy-flux pair for the proposed system and provided there exists $\delta > 0$ such that

$$\operatorname{sgn}(\gamma) f'(u) > \delta$$

3 - Entropy / Entropy-flux pair - proof elements

Identify entropy variables in the system :

$$E^\alpha(\mathbf{U}) = \text{sgn}(\gamma)F(u) + \frac{|\gamma|}{2}p^2 + \frac{1}{2\alpha}\psi^2 + \frac{\beta}{2}w^2,$$

$$u_t + (f(u) + \text{sgn}(\gamma)\psi)_x = 0,$$

$$\psi_t + \alpha(f(u) + \text{sgn}(\gamma)\psi)_x = -\alpha w,$$

$$w_t - v_\beta^2 p_x = \beta^{-1} \psi,$$

$$p_t - w_x = 0$$

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$$\begin{aligned} u_t + (f(u) + \operatorname{sgn}(\gamma)\psi)_x &= 0, & u_t + \operatorname{sgn}(\gamma)(E_u + \alpha E_\psi)_x &= 0, \\ \psi_t + \alpha(f(u) + \operatorname{sgn}(\gamma)\psi)_x &= -\alpha w, & \psi_t + \alpha \operatorname{sgn}(\gamma)(E_u + \alpha E_\psi)_x &= -\alpha\beta^{-1}E_w, \\ w_t - v_\beta^2 p_x &= \beta^{-1}\psi, & w_t - (\beta^{-1}E_p)_x &= \alpha\beta^{-1}E_\psi, \\ p_t - w_x &= 0 & p_t - (\beta^{-1}E_w)_x &= 0 \end{aligned} \quad \Rightarrow$$

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Thus, multiplying by E_u, E_ψ, E_w, E_p and summing all equations

$$E_t + \left(\operatorname{sgn}(\gamma) \frac{1}{2} E_u^2 + \operatorname{sgn}(\gamma) \alpha E_u E_\psi + \frac{\alpha^2}{2} E_\psi^2 - \beta^{-1} E_p E_w \right)_x = 0$$

4 - Symmetrizability and local well-posedness 1/2

For $\alpha > \max |f'(u)|$, the hyperbolic system can be cast into a symmetric t -hyperbolic form of Friedrichs.

4 - Symmetrizability and local well-posedness 1/2

For $\alpha > \max |f'(u)|$, the hyperbolic system can be cast into a symmetric t -hyperbolic form of Friedrichs. Indeed consider the map

$$\mathcal{T} : \mathbb{R}^4 \mapsto \mathbb{R}^4, \quad \mathbf{U} = \begin{pmatrix} u \\ \psi \\ w \\ p \end{pmatrix} \mapsto \Phi = \begin{pmatrix} \phi_1 \\ \phi_2 \\ \phi_3 \\ \phi_4 \end{pmatrix}, \quad \text{with} \quad \begin{cases} \phi_1 = \operatorname{sgn}(\gamma) f(u) + \psi, \\ \phi_2 = \alpha u - \psi, \\ \phi_3 = v_\beta p, \\ \phi_4 = w. \end{cases}$$

The Jacobian of this transformation is given by

$$D\mathcal{T}(\mathbf{U}) = \begin{pmatrix} \operatorname{sgn}(\gamma) f'(u) & 1 & 0 & 0 \\ \alpha & -1 & 0 & 0 \\ 0 & 0 & 0 & v_\beta \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad \det(D\mathcal{T}(\mathbf{U})) = v_\beta (\alpha + \operatorname{sgn}(\gamma) f'(u)) \neq 0$$

4 - Symmetrizability and local well-posedness 2/2

- Under this change of variables the system becomes

$$\Phi_t + \tilde{\mathbf{A}}(\Phi)\Phi_x = \tilde{\mathbf{b}}(\Phi),$$

with the symmetric matrix

$$\tilde{\mathbf{A}}(\Phi) = \begin{pmatrix} \operatorname{sgn}(\gamma)\alpha + f'(u(\Phi)) & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -v_\beta \\ 0 & 0 & -v_\beta & 0 \end{pmatrix}.$$

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⇒ Friedrichs-symmetrizable

4 - Symmetrizability and local well-posedness 2/2

- Under this change of variables the system becomes

$$\Phi_t + \tilde{\mathbf{A}}(\Phi)\Phi_x = \tilde{\mathbf{b}}(\Phi),$$

with the symmetric matrix

$$\tilde{\mathbf{A}}(\Phi) = \begin{pmatrix} \operatorname{sgn}(\gamma)\alpha + f'(u(\Phi)) & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -v_\beta \\ 0 & 0 & -v_\beta & 0 \end{pmatrix}.$$

⇒ Friedrichs-symmetrizable ⇒ the Cauchy problem is locally well-posed for

$$\mathbf{U}_0 \in (H^s(I))^4, \quad s > 3/2.$$

5 - Riemann invariants

The system possesses a full set of Riemann invariants, which form a coordinate system. The Riemann invariants $R_1 = R_1(\mathbf{U}), \dots, R_4 = R_4(\mathbf{U})$ are given by

$$R_1 = w + v_\beta p, \quad R_2 = \alpha u - \psi, \quad R_3 = w - v_\beta p, \quad R_4 = \operatorname{sgn}(\gamma) f(u) + \psi.$$

Indeed, we have

$$\ell_i = \ell_i(\mathbf{U}) = (R_{i,u}(\mathbf{U}), R_{i,\psi}(\mathbf{U}), R_{i,w}(\mathbf{U}), R_{i,p}(\mathbf{U})) \quad (i = 1, \dots, 4, \mathbf{U} \in \mathbb{R}^4),$$

are each a left eigenvector associated with λ_i .

Summary so far

We have approximated the purely-dispersive equation

$$u_t + f(u)_x = \gamma u_{xxx}$$

via the first-order system

$$\begin{aligned} u_t^\alpha + f(u^\alpha)_x + \psi_x &= 0, \\ \psi_t^\alpha + \alpha f(u^\alpha)_x + \alpha \psi_x &= -\alpha w^\alpha, \\ \beta w_t^\alpha - \gamma p_x^\alpha &= \psi, \\ p_t^\alpha - w_x^\alpha &= 0. \end{aligned}$$

We have shown

- ① Hyperbolicity, Symmetrizability & Local well-posedness
- ② Existence of Riemann invariants
- ③ Derivation from a least-action principle
- ④ Existence of an Entropy/Entropy-flux pair

Dissipative extension possibilities

- 1 Simply insert the parabolic term back to the system

$$\begin{aligned} u_t^\alpha + (f(u^\alpha) + \operatorname{sgn}(\gamma)\psi^\alpha)_x &= \varepsilon u_{xx}^\alpha, \\ \psi_t^\alpha + \alpha (f(u^\alpha) + \operatorname{sgn}(\gamma)\psi^\alpha)_x &= -\alpha w^\alpha, \\ w_t^\alpha - \beta^{-1} |\gamma| p_x^\alpha &= \beta^{-1} \psi^\alpha, \\ p_t^\alpha - w_x^\alpha &= 0 \end{aligned}$$

Dissipative extension possibilities

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- 2 Derive the energy equation

$$\begin{aligned} E_t^\alpha + \left(\frac{1}{2} f(u^\alpha)^2 + f(u^\alpha)\psi + \operatorname{sgn}(\gamma) \frac{\psi^{\alpha 2}}{2} - |\gamma| p^\alpha w^\alpha \right)_x &= \varepsilon f(u^\alpha) u_{xx}^\alpha. \\ &= (\varepsilon f(u^\alpha) u_x^\alpha)_x - \varepsilon f'(u^\alpha) (u_x^\alpha)^2 \end{aligned}$$

Dissipative extension possibilities

- 1 Simply insert the parabolic term back to the system

$$\begin{aligned} u_t^\alpha + (f(u^\alpha) + \operatorname{sgn}(\gamma)\psi^\alpha)_x &= \varepsilon u_{xx}^\alpha, \\ \psi_t^\alpha + \alpha (f(u^\alpha) + \operatorname{sgn}(\gamma)\psi^\alpha)_x &= -\alpha w^\alpha, \\ w_t^\alpha - \beta^{-1} |\gamma| p_x^\alpha &= \beta^{-1} \psi^\alpha, \\ p_t^\alpha - w_x^\alpha &= 0 \end{aligned}$$

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Dissipated entropy for increasing flux functions $f(u)$ (convex Energy $F(u)$)

Second possibility: Visco-elastic relaxation

Introduce the following relaxed system

$$u_t + f(u)_x + \operatorname{sgn}(\gamma)\alpha(u - c)_x = \varepsilon u_{xx},$$

$$\beta(c_{tt} - \varepsilon c_{xxt}) - |\gamma|c_{xx} = \alpha(u - c)$$

Second possibility: Visco-elastic relaxation

Introduce the following relaxed system

$$u_t + f(u)_x + \operatorname{sgn}(\gamma)\alpha(u - c)_x = \varepsilon u_{xx},$$

$$\beta(c_{tt} - \varepsilon c_{xxt}) - |\gamma|c_{xx} = \alpha(u - c)$$

Define the variables

$$w = c_t - \varepsilon c_{xx}, \quad p = c_x, \quad \psi = \alpha(u - c),$$

Obtain

$$u_t + (f(u) + \operatorname{sgn}(\gamma)\psi)_x = \varepsilon u_{xx},$$

$$\psi_t + \alpha(f(u) + \operatorname{sgn}(\gamma)\psi)_x = -\alpha w + \varepsilon \psi_{xx},$$

$$w_t - \beta^{-1}|\gamma|p_x = \beta^{-1}\psi,$$

$$p_t - w_x = \varepsilon p_{xx}$$

Second possibility: Visco-elastic relaxation

$$\frac{d}{dt} \int_I E^\alpha(\mathbf{U}) dx = -\varepsilon \int_I \left(\operatorname{sgn}(\gamma) f'(u) u_x^2 + |\gamma| p_x^2 + \frac{\psi_x^2}{\alpha} \right) dx \leq 0.$$

Again, entropy stable for $f(u)$ increasing in u .

Second possibility: Visco-elastic relaxation

$$\frac{d}{dt} \int_I E^\alpha(\mathbf{U}) dx = -\varepsilon \int_I \left(\operatorname{sgn}(\gamma) f'(u) u_x^2 + |\gamma| p_x^2 + \frac{\psi_x^2}{\alpha} \right) dx \leq 0.$$

Again, entropy stable for $f(u)$ increasing in u .

Compatible with **one of the** original system entropies. Indeed

$$u_t + f(u)_x = \varepsilon u_{xx} + \gamma u_{xxx}$$

admits (among others) the entropy

$$E[u] = \operatorname{sgn}(\gamma) F(u) + \frac{|\gamma| u_x^2}{2}$$

satisfying

$$\frac{d}{dt} \int_I E[u] dx = -\varepsilon \int_I \left((\operatorname{sgn}(\gamma) f'(u) u_x^2 + |\gamma| u_{xx}^2) \right) dx$$

Convergence of solutions (also works for $\varepsilon = 0$)

Theorem (I)

Let $u_0 \in H^6(I)$, and suppose that $|f'(u)| < L$. We assume that $u \in L^2(0, T; H^3(I))$ with $u_{tt} \in L^2(I_T)$ and $\|u\|_{L^\infty(0, T; H^4(I))} < \infty$ is a classical solution of

$$u_t + f(u)_x = \varepsilon u_{xx} + \gamma u_{xxx}, \quad u(x, 0) = u_0(x)$$

Consider for $\beta = \alpha^{-1}$, a sequence $\{\mathbf{U}^\alpha = (u^\alpha, \psi^\alpha, w^\alpha, p^\alpha)^\top\}_{\alpha > 0}$ of weak entropy solutions of

$$\mathbf{U}_t^\alpha + \mathbf{f}(\mathbf{U}^\alpha)_x = \mathbf{S}(\mathbf{U}^\alpha) + \mathbf{D}\mathbf{U}^\alpha_{xx} \quad + \quad \text{Well-prepared IC}$$

Then there is a constant $C = C(L, \delta, u) \geq 0$ that is independent of α such that

$$\|u^\alpha - u\|_{L^\infty(0, T; L^2(I))} + \|p^\alpha - u_x\|_{L^\infty(0, T; L^2(I))} \leq C\alpha^{-1/2}.$$

Convergence of solutions - proof elements

- 1 Write the original system in a similar form as its approximation.

$$\begin{aligned}
 u_t + f(u)_x + \operatorname{sgn}(\gamma)\psi_x &= 0, \\
 \frac{1}{\alpha}\psi_t + f(u)_x + \operatorname{sgn}(\gamma)\psi_x &= -w - \frac{1}{\alpha}|\gamma|u_{xxt}, \\
 \beta w_t - |\gamma|p_x &= \psi + \beta u_{tt}, \\
 p_t - w_x &= 0.
 \end{aligned} \tag{4}$$

- 2 Use Relative entropy framework.
- 3 Grönwall type arguments.

Global well-posedness of the relaxation system

Theorem (II)

Let $\alpha, \beta > 0$ and $\gamma \neq 0$. Suppose the flux f satisfies Assumption $\mathcal{A}$ (monotonicity). Then there exists $\Lambda > 0$ such that for all $\mathbf{U}_0 = (u_0, \psi_0, w_0, p_0)^\top \in (H^s(I))^4$, $s \geq 2$, satisfying

$$\|u_0 - u^*\|_{H^s}^2 + \frac{1}{\alpha} \|\psi_0\|_{H^s}^2 + \beta \|w_0\|_{H^s}^2 + |\gamma| \|p_0\|_{H^s}^2 < \Lambda,$$

the IVP for the relaxation system admits a unique classical solution

$$\mathbf{U} - \mathbf{U}^* \in C([0, \infty); (H^s(I))^4) \cap C^1([0, \infty); (H^{s-1}(I))^4).$$

Moreover,

$$\sup_{t \geq 0} \mathcal{E}_s(t) + \varepsilon \int_0^t \mathcal{D}_s(\tau) d\tau \leq C(\Lambda) \mathcal{E}_s(0),$$

where $\mathcal{E}_s$ denotes the H^s energy.

Global well-posedness – proof elements

- 1 Use the entropy $E^\alpha(\mathbf{U})$ as a symmetrizer:

$$\mathbf{S}_0(\mathbf{U})\mathbf{U}_t + \mathbf{S}_1(\mathbf{U})\mathbf{U}_x - \mathbf{S}_2(\mathbf{U})\mathbf{U}_{xx} = \mathbf{S}_3(\mathbf{U})\mathbf{U}.$$

- 2 The system becomes a **symmetric hyperbolic–parabolic system**.
- 3 Verify the **Shizuta–Kawashima coupling condition**:

$$\ker \mathbf{S}_2 \cap \ker(\mathbf{S}_1 - \lambda I) = \{0\}.$$

- 4 The source matrix is **skew-symmetric**, hence it does not contribute to the energy:

$$\partial_x^s(\mathbf{U} - \mathbf{U}^*)^\top \mathbf{S}_3 \partial_x^s(\mathbf{U} - \mathbf{U}^*) = 0.$$

- 5 Apply the classical theory of Kawashima (1984) for partially diffusive systems.

1 The hyperbolic approximation

2 Model properties

3 Numerical results

Semi-Implicit Lax Wendroff - Intercell values

In order to solve the hyperbolic-parabolic system

$$\mathbf{U}_t^\alpha + \mathbf{f}(\mathbf{U}^\alpha)_x = \mathbf{S}(\mathbf{U}^\alpha)\mathbf{U}^\alpha + \mathbf{D}\mathbf{U}_{xx}^\alpha$$

We use a semi-implicit Richtmeyer-Lax-Wendroff method.

We first evaluate the intercell values $\mathbf{U}_{i\pm\frac{1}{2}}^{n+\frac{1}{2}}$ via

$$\begin{aligned}\mathbf{U}_{i+\frac{1}{2}}^{n+\frac{1}{2}} &= \frac{1}{2} (\mathbf{U}_i^n + \mathbf{U}_{i+1}^n) - \frac{\Delta t}{2\Delta x} (\mathbf{f}(\mathbf{U}_{i+1}^n) - \mathbf{f}(\mathbf{U}_i^n)) + \frac{\Delta t}{2} \mathbf{S}\mathbf{U}_{i+\frac{1}{2}}^{n+\frac{1}{2}}, \\ \mathbf{U}_{i-\frac{1}{2}}^{n+\frac{1}{2}} &= \frac{1}{2} (\mathbf{U}_i^n + \mathbf{U}_{i-1}^n) - \frac{\Delta t}{2\Delta x} (\mathbf{f}(\mathbf{U}_i^n) - \mathbf{f}(\mathbf{U}_{i-1}^n)) + \frac{\Delta t}{2} \mathbf{S}\mathbf{U}_{i-\frac{1}{2}}^{n+\frac{1}{2}},\end{aligned}$$

whose values can be computed explicitly as

$$\mathbf{U}_{i\pm\frac{1}{2}}^{n+\frac{1}{2}} = \left(\mathbf{I} - \frac{\Delta t}{2} \mathbf{S} \right)^{-1} \left(\frac{1}{2} (\mathbf{U}_i^n + \mathbf{U}_{i\pm 1}^n) \mp \frac{\Delta t}{2\Delta x} (\mathbf{f}(\mathbf{U}_{i\pm 1}^n) - \mathbf{f}(\mathbf{U}_i^n)) \right).$$

Semi-Implicit Lax Wendroff - Update and parabolic terms

The solution of the hyperbolic first-order part of the system is then given by

$$\mathbf{U}_i^* = \mathbf{U}_i^n - \frac{\Delta t}{\Delta x} \left(\mathbf{f} \left(\mathbf{U}_{i+\frac{1}{2}}^{n+\frac{1}{2}} \right) - \mathbf{f} \left(\mathbf{U}_{i-\frac{1}{2}}^{n+\frac{1}{2}} \right) \right) + \frac{\Delta t}{2} \mathbf{S} \left(\mathbf{U}_{i+\frac{1}{2}}^{n+\frac{1}{2}} + \mathbf{U}_{i-\frac{1}{2}}^{n+\frac{1}{2}} \right).$$

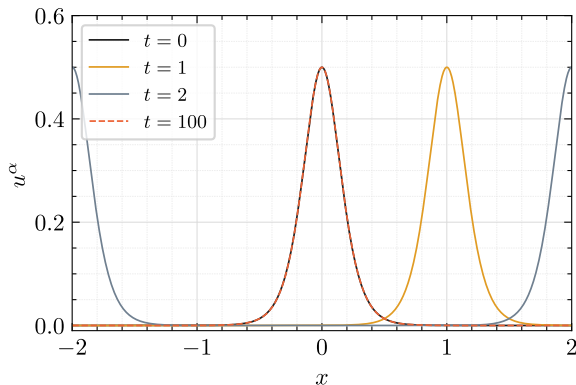
Finally, the solution at the new time, accounting for the second-order terms (present for $\varepsilon > 0$) is

$$\mathbf{U}_i^{n+1} = \mathbf{U}_i^* + \frac{\Delta t}{\Delta x^2} \mathbf{D} \left(\mathbf{U}_{i+1}^* - 2\mathbf{U}_i^* + \mathbf{U}_{i-1}^* \right).$$

KdV 1-Soliton solution

Target problem

$$\begin{cases} u_t + 6uu_x = \gamma u_{xxx}, \\ u(x, 0) = \frac{V}{2} \operatorname{sech}^2 \left(\sqrt{\frac{V}{4|\gamma|}} x \right), \\ \text{Periodic BC.} \end{cases}$$



$I_c = [-2, 2]$ with $N = 1000$ cells, $\gamma = -10^{-2}$, $\alpha = 10^3$ and $\beta = 10^{-6}$. Wave velocity $V = 1$ and final time is $T = 100$ (25 periods).

KdV 2-Soliton solution

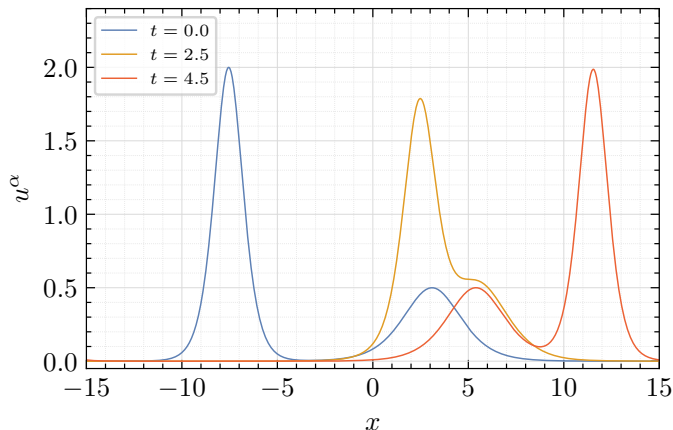
Target problem

$$\left\{ \begin{array}{l} u_t + 6uu_x = -u_{xxx}, \\ u(x, 0) = \frac{2(V_1 - V_2) (V_1 \cosh^2 \theta_2(x) + V_2 \sinh^2 \theta_1(x))}{((\sqrt{V_1} + \sqrt{V_2}) \cosh(\theta_1(x) - \theta_2(x)) + (\sqrt{V_1} - \sqrt{V_2}) \cosh(\theta_1(x) + \theta_2(x)))^2}, \\ \text{where } \theta_j(x) = \sqrt{V_j}(x - x_j)/2, \quad j = 1, 2 \\ \text{Periodic BC.} \end{array} \right.$$

KdV 2-Soliton solution

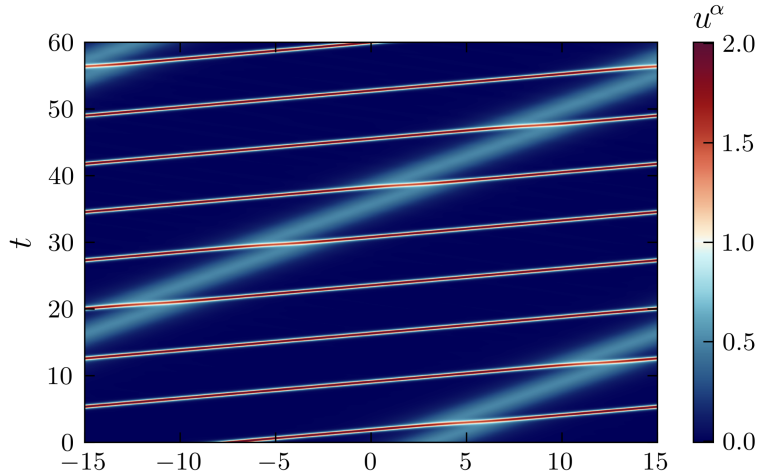
Target problem

$$\begin{cases} u_t + 6uu_x = -u_{xxx}, \\ u(x, 0) = \dots \\ \text{Periodic BC.} \end{cases}$$



$I_c = [-15, 15]$ with $N = 1000$ cells, $\alpha = 4000$ and $\beta = 10^{-6}$. Final time is $T = 60$.

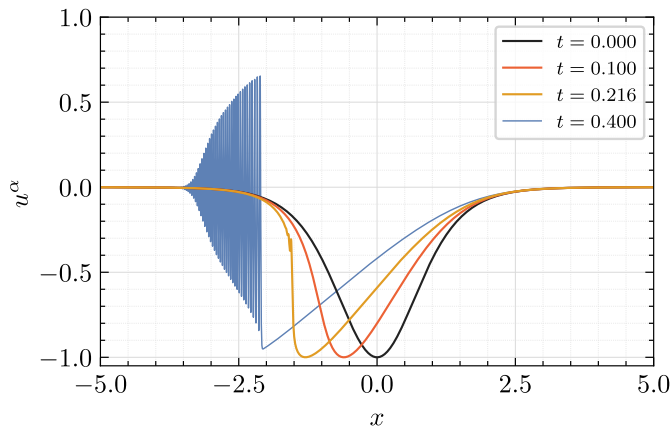
KdV 2-Soliton solution $(x - t)$ diagram



KdV negative hump DSW

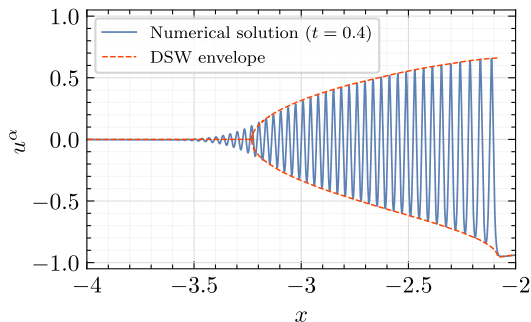
Target problem

$$\begin{cases} u_t + 6uu_x = \gamma u_{xxx}, \\ u(x, 0) = -\text{sech}^2(x), \\ \text{Periodic BC.} \end{cases}$$

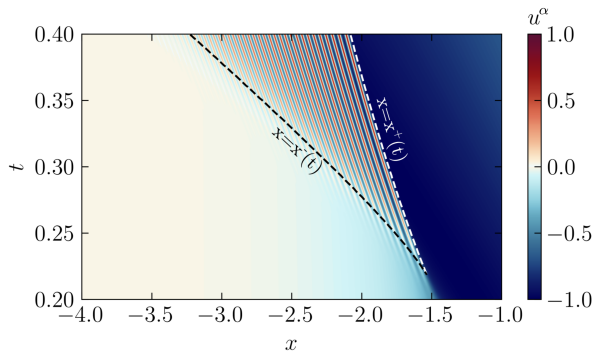


$I_c = [-5, 5]$ with $N = 8000$ cells, $\gamma = -10^{-4}$, $\alpha = 10^3$ and $\beta = 10^{-7}$. Final time is $T = 0.4$.

KdV negative hump DSW



(a) Comparison of the numerical solution with the Whitham envelope for the target problem.



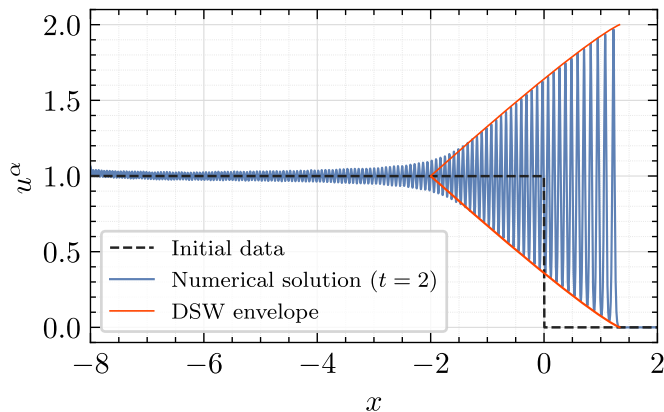
(b) $x - t$ diagram of the numerical solution at the onset of the DSW formation.

$I_c = [-5, 5]$ with $N = 8000$ cells, $\gamma = -10^{-4}$, $\alpha = 10^3$ and $\beta = 10^{-7}$. Final time is $T = 0.4$.

KdV Riemann problem

Target problem

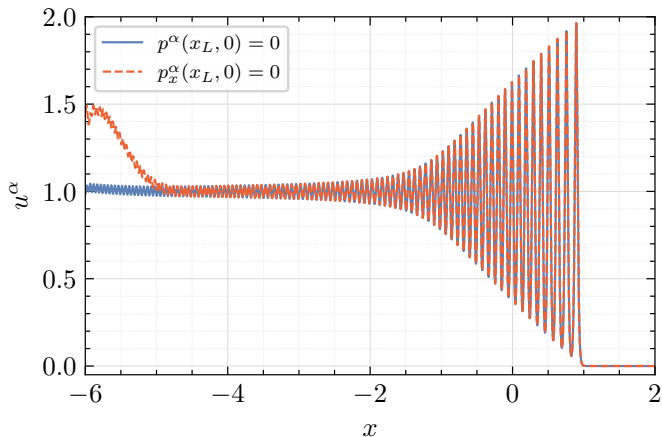
$$\begin{cases} u_t + uu_x = \gamma u_{xxx}, \\ u(x, 0) = \begin{cases} 1 & \text{if } x < 0, \\ 0 & \text{if } x \geq 0. \end{cases} \end{cases}$$



$I_c = [-8, 2]$ with $N = 10000$ cells, $\gamma = -10^{-4}$, $\alpha = 10^3$ and $\beta = 10^{-7}$. Final time is $T = 0.4$.

"Homogeneous Neumann" Boundary conditions

$$\begin{aligned} u_{\mathbf{x}}(x_L, t) &= u_{\mathbf{x}}(x_R, t) = 0, \\ \psi_{\mathbf{x}}(x_L, t) &= \psi_{\mathbf{x}}(x_R, t) = 0, \\ w_{\mathbf{x}}(x_L, t) &= w_{\mathbf{x}}(x_R, t) = 0, \\ p(x_L, t) &= p(x_R, t) = 0. \end{aligned}$$



Comparison of suggested "Neumann" boundary conditions with classic Neumann boundary

KdV-Burgers Traveling wave front

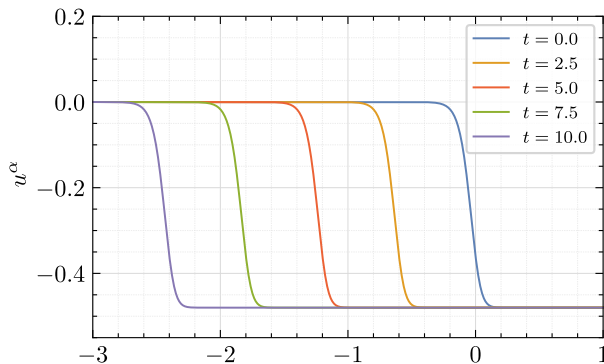
Target problem

$$\begin{cases} u_t + uu_x = \varepsilon u_{xx} + \gamma u_{xxx}, \\ u(x, 0) = \frac{-3\varepsilon^2}{25\gamma} \left(2 + 2 \tanh \left(\frac{\varepsilon x}{10\gamma} \right) + \operatorname{sech}^2 \left(\frac{\varepsilon x}{10\gamma} \right) \right). \end{cases}$$

KdV-Burgers Traveling wave front

Target problem

$$\begin{cases} u_t + uu_x = \varepsilon u_{xx} + \gamma u_{xxx}, \\ u(x, 0) = \dots \end{cases}$$



$I_c = [-3, 1]$ with $N = 2000$ cells, $\gamma = 10^{-4}$, $\alpha = 10^4$ and $\beta = 10^{-6}$. Solution is shown at different times up to $T = 10$.

Convergence in the relaxation parameters

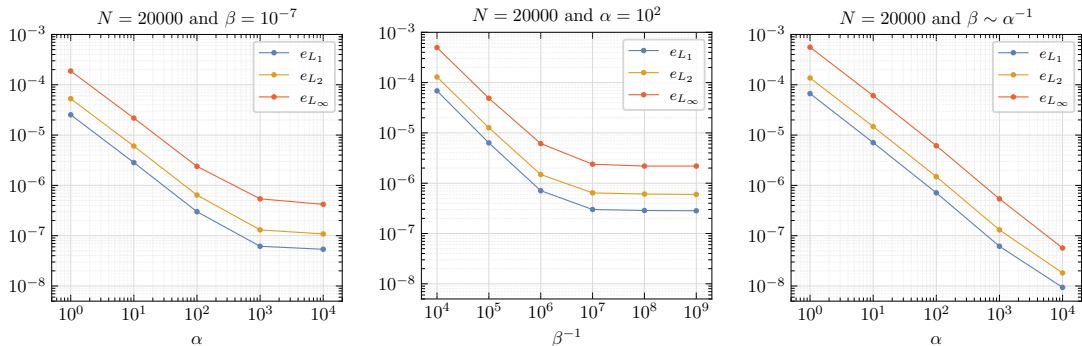


Figure 7: Convergence plots in different error norms. Left: Error decay and saturation for fixed β and $\alpha \rightarrow \infty$. Middle: Error decay and saturation for fixed β and $\alpha \rightarrow \infty$. Right: Linear error decay for $\beta = \gamma\alpha^{-1}$ and $\alpha \rightarrow \infty$.

mKdV-Burgers Undercompressive shock

$$\begin{cases} u_t + 3u^2u_x = \varepsilon u_{xx} - \gamma u_{xxx}, \\ u(x, 0) = \frac{1}{2} (u_+ + u_- - |u_+ - u_-| \tanh(Ax/\sqrt{\gamma})) , \end{cases}$$

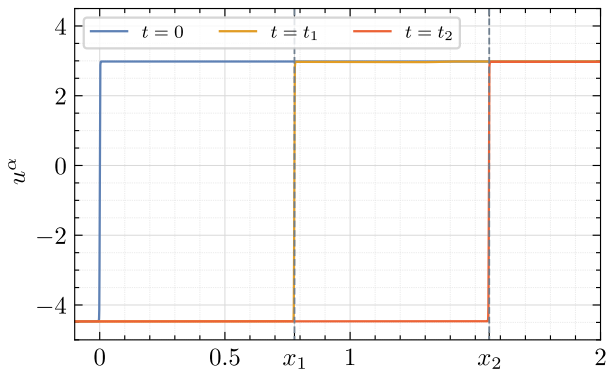
$$u_- = -\sqrt{\frac{2\varepsilon^2}{\gamma}}, \quad u_+ = -u_- - \sqrt{\frac{2\varepsilon^2}{9\gamma}}, \quad A = \frac{u_- - u_+}{2\sqrt{2}}.$$

For these values, in the zero dispersion/dissipation limit, an undercompressive shock is obtained since the corresponding shock is subsonic both ahead of and behind the wave:

$$\begin{aligned} u_- &\simeq -4.47, & u_+ &\simeq 2.97, & s &= \frac{u_+^3 - u_-^3}{u_+ - u_-} \\ \lambda_- &\simeq 59.94, & \lambda_+ &\simeq 26.46, & s &\simeq 15.52 \end{aligned}$$

parameters: $\varepsilon = 10^{-2}$, $\gamma = 10^{-5}$

mKdV-Burgers Undercompressive shock



$I_c = [-0.1, 2]$ with $N = 10000$ cells, $\gamma = 10^{-5}$, $\alpha = 10^3$ and $\beta = 10^{-11}$. Solution is shown at different times up to $T = 10$.

Gardner equation example

We consider the Gardner equation given by

$$u_t + 6(u - ku^2)u_x + u_{xxx} = 0, \quad u(x, 0) = u_{d,b}(x), \quad (5)$$

where the initial conditions $u_d(x)$ and $u_b(x)$, corresponding to dark or bright solitons, respectively

$$u_d(x) = u_3 - \frac{u_3 - u_2}{\cosh^2(\theta(x)) - \frac{u_3 - u_2}{u_3 - u_1} \sinh^2(\theta(x))},$$

$$u_b(x) = u_2 + \frac{u_3 - u_2}{\cosh^2(\theta(x)) - \frac{u_3 - u_2}{u_4 - u_2} \sinh^2(\theta(x))}.$$

Here $u_1 \leq u_2 \leq u_3 \leq u_4$ and $\theta(x) = \frac{1}{2} \sqrt{k(u_3 - u_1)(u_4 - u_2)}x$. The exact solution to (5) is then given by $u_{b,d}(x - Vt)$ where the velocity V is given as

$$V = k(u_1u_2 + u_1u_3 + u_1u_4 + u_2u_3 + u_2u_4 + u_3u_4). \quad (6)$$

Gardner equation - parameters

we consider a small parameter $\epsilon > 0$, and we take for the dark soliton solution

$$k = 1, \quad u_1 = 0, \quad u_2 = \epsilon, \quad u_3 = u_4 = 1 - \epsilon,$$

and analogously for the bright soliton

$$k = 1, \quad u_1 = u_2 = \epsilon, \quad u_3 = 1 - \epsilon, \quad u_4 = 1.$$

This results in table-top-like solitons for $\epsilon \ll 1$. For both parameterizations, we take $\epsilon = 10^{-4}$ and the value of the traveling wave velocity is $V = 1 - \epsilon^2 \simeq 1$.

Gardner equation

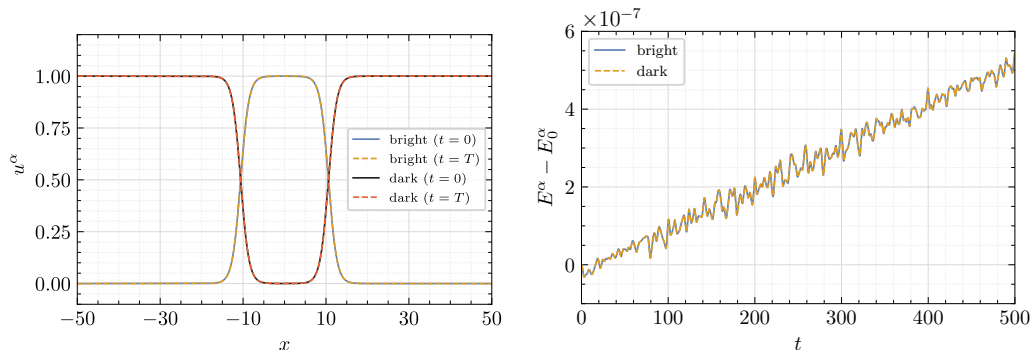


Figure 8: Left: Numerical solutions to the AIVP corresponding to (5) for both bright and dark solitons. Right: Evolution of the total energy error $E^\alpha - E_0^\alpha$ over time.

$I_c = [-50, 50]$, $N = 2000$, $\alpha = 10^3$, $\beta = 10^{-6}$ and $T = 500/V$.

Conclusions - Perspectives

Summary

- ① A new first-order hyperbolic model for general conservation laws with dissipative/dispersive regularization is presented.
- ② Convergence of its solutions towards those of the original model is proven for sufficiently regular solutions.
- ③ The numerical results show good performance of the approach for several PDEs.

Perspectives

- ① Performance - comparison with existing methods ? Is it worth it ?
- ② proper implementation of boundary conditions.
- ③ Fully hyperbolic model ?
- ④ Vanishing-viscosity solution convergence for hyperbolized system ?



Thank you for your attention !

[1] Barthwal, Rahul, Firas Dhaouadi, and Christian Rohde. "On hyperbolic approximations for a class of dispersive and diffusive-dispersive equations." arXiv preprint arXiv:2512.04882 (2025).

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